



Topics in this chapter:

1. Acids, Bases, Salts & pH
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3. Metals, Non-metals & Reactivity Series
4. Ionic vs Covalent Compounds
5. Redox, Corrosion & Rancidity
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1. Acids, Bases, Salts & pH

- Acid + Base → Salt + Water (neutralization). [Q1353, Q1378]
- Sodium bicarbonate (NaHCO_3 , Baking Soda):
 - Antacid — excess stomach acid neutralize karta hai.
 - HCl ke saath react karke CO_2 gas deta hai: $\text{NaHCO}_3 + \text{HCl} \rightarrow \text{NaCl} + \text{CO}_2 + \text{H}_2\text{O}$.
 - Bee sting (formic acid HCOOH) ko neutralize karne ke liye use hota hai.
 - Solvay process se industrially produce hota hai.
 - Fire extinguishers, cosmetics, ear drops mein bhi use hota hai. [Q1353, Q1378]
- Sodium carbonate (Na_2CO_3 , Washing Soda):
 - Hard water ko normalize karta hai — temporary aur permanent dono hardness remove karta hai ($\text{Ca}^{2+}/\text{Mg}^{2+}$ ions ko insoluble carbonate mein precipitate karta hai).
 - Detergents, soaps, paper, water glass, borax manufacturing mein use hota hai.
 - Stain removal ke liye bhi use hota hai. [Q1351, Q1367, Q1396]
- Sodium hydroxide (NaOH , Caustic Soda / Lye):
 - Soaps, rayon, paper, explosives, dyestuffs, petroleum products, cotton processing, metal cleaning, electroplating mein use hota hai.
 - Permanent hardness remove karne ke liye use NAHI hota — ye uska use nahi hai. [Q1354]
- Alum (Fitkari): Double salt ka example — equimolar proportions of $\text{K}_2\text{SO}_4 + \text{Al}_2(\text{SO}_4)_3$ crystallise karke banta hai. Mohr's salt ($\text{FeSO}_4 \cdot (\text{NH}_4)_2\text{SO}_4 \cdot 6\text{H}_2\text{O}$) bhi double salt hai. [Q1383]
- Bleaching Powder $\{\text{Ca}(\text{ClO})_2 / \text{CaOCl}_2\}$: Oxidising agent hai; chlorine gas + calcium hydroxide se banta hai; water purification, textile, paper industries mein use hota hai. [Q1375]
- pH Scale: Acidic < 7; Neutral = 7; Basic > 7. [Q1416]
- Amphoteric oxides: Al_2O_3 , ZnO — acid aur base donon se react karte hain. [Q1372]
- Basic oxides: Metals ke oxides jo water se basic solution banate hain — e.g., Na_2O (sodium oxide, Na = atomic

number 11). [Q1393]

- CaO (Quicklime): White, caustic, alkaline, crystalline solid at room temperature. [Q1361]
- Ca(OH)₂ (Slaked Lime): Soft white powder; CO₂ se react karke limestone (CaCO₃) banata hai. [Q1361]
- Denatured alcohol: Ethanol + poisonous substances (methanol) + blue dye — peene ke liye unfit; easily recognized. [Q1395]
- Pb(NO₃)₂ + 2KI → PbI₂↓ (yellow precipitate) + 2KNO₃ — double displacement reaction jismein lead aur potassium apne anions exchange karte hain. [Q1356]

2. Important Compounds & Their Uses

- Plaster of Paris (POP):
 - Calcium sulphate hemihydrate: CaSO₄·½H₂O.
 - Gypsum (CaSO₄·2H₂O) ko 373 K (100–120°C) par heat karke banta hai.
 - Water ke saath hard mass mein set hota hai.
 - Uses: False ceiling, sculptures, moulds, bone fracture plasters, blackboard chalk, dentistry, fireproofing, chemical lab mein airtight sealing.
 - Moisture se bachake sealed container mein rakhna chahiye. [Q1352, Q1355, Q1369]
- Gypsum: Calcium sulphate dihydrate (CaSO₄·2H₂O). [Q1355]
- Limestone (CaCO₃): Sedimentary rock; cement, building construction, chalk, crushed stone mein use hota hai. [Q1368]
- Calcium carbide (CaC₂): Acetylene aur calcium cyanamide production mein. [Q1361]
- Limewater test: CO₂ + Ca(OH)₂ → CaCO₃↓ (white ppt) + H₂O; excess CO₂ se Ca(HCO₃)₂ (soluble) banta hai. [Q1407]
- Ammonia (NH₃):
 - Colourless, sharp suffocating odour.
 - N:H mass ratio = 14:3.
 - ~80% fertilizer ke roop mein use hota hai; refrigerant, water purification, plastics, explosives mein bhi. [Q1414]
 - ⚠ Trap: NH₃ acid rain se related NAHI hai — ye nutritious, inorganic nitrogen compound hai. [Q1371]
- Chlorine (Cl₂): Greenish-yellow, toxic, corrosive, irritating to eyes/respiratory system. [Q1340]
- MgCl₂: Highly soluble in water; insoluble in acetone. Hydrates form karta hai (MgCl₂·nH₂O, n=1 to 12). [Q1384]
- CaCl₂: White crystalline solid, highly soluble in water; ionic compound with very high boiling point (strong electrostatic forces). [Q1365]

3. Metals, Non-metals & Reactivity Series

- Reactivity series (high to low): K > Na > Ca > Mg > Al > Zn > Fe > Pb > H > Cu > Hg > Ag > Au. [Q1364, Q1381]
- Metals: Electropositive nature — 1, 2, 3 valence electrons; easily lose karke positive ions banate hain. [Q1380]
- Non-metals: Electronegative nature — electrons gain karke negative ions banate hain. [Q1380]
- Sodium (Na): Atomic number 11; Group 1; highly reactive — oxygen aur moisture se vigorously react karta hai; isliye kerosene mein store karte hain. [Q1376]
- Fluorine (F): Sabse zyada reactive halogen; atomic number 9. Halogens ki reactivity upar se neeche jaate hue kam hoti hai. [Q1377]
- Displacement reaction: Zyada reactive metal kam reactive metal ke salt solution se usko displace kar sakta hai.
 - Zn + CuSO₄ → ZnSO₄ + Cu (Zn is more reactive than Cu). [Q1381]
 - Fe + CuSO₄ → FeSO₄ + Cu (Fe surface brown copper coating ban jaata hai; solution blue se pale green ho jaata)

hai). [Q1364]

- $\text{Zn} + \text{H}_2\text{SO}_4 \rightarrow \text{ZnSO}_4 + \text{H}_2\uparrow$: H_2 gas colourless hai aur pop sound ke saath jalta hai. [Q1359]
- White phosphorus: Highly reactive with oxygen; isliye water mein store karte hain. Translucent white waxy solid; burning se dense white irritating smoke (phosphorus oxides) nikalta hai. [Q1382]
- Halogens: Group 17; valency = 1; reactivity, electronegativity down the group kam hoti hai; melting/boiling points, atomic radius badhti hai. [Q1377]

4. Ionic vs Covalent Compounds

- Ionic compounds:
 - Electron transfer se bante hain; cation + anion.
 - Properties: Solid, hard, high melting & boiling points (strong electrostatic forces), soluble in water, insoluble in petrol, conduct electricity in aqueous/molten state.
 - Examples: NaCl, CaCl_2 , MgCl_2 , NaBr, KBr. [Q1365, Q1397]
- Covalent compounds:
 - Covalent bond se bante hain.
 - Properties: Low melting & boiling points, weak conductors of heat/electricity, insoluble in water, soluble in organic solvents.
 - Examples: Glucose, alcohol, kerosene, octane. [Q1386, Q1387]
- ⚠ Trap: High melting & boiling points covalent compound ka property NAHI hai — ye ionic ka property hai. [Q1386]
- ⚠ Trap: Low melting & boiling points ionic compound ka property NAHI hai. [Q1397]
- Sugar solution electricity conduct nahi karta kyunki sugar molecules ions mein dissociate nahi hote. [Q1387]
- NaCl(aq) electricity conduct karta hai kyunki Na^+ aur Cl^- ions free hote hain. [Q1366]

5. Redox, Corrosion & Rancidity

- Oxidation: Oxygen gain ya hydrogen loss. [Q1360]
- Reduction: Oxygen loss ya hydrogen gain. [Q1360]
- Rusting: $4\text{Fe} + 3\text{O}_2 \rightarrow 2\text{Fe}_2\text{O}_3$; Fe oxidized hota hai. [Q1360]
- Corrosion: Metal surface ka air/moisture se damage — iron rust, copper green coating, silver black.
 - Prevention: Painting, oiling, greasing, galvanising, chrome plating. [Q1390]
- Rancidity: Fats/oils ka oxidation/hydrolysis se spoilage — unpleasant odour; antioxidants, airtight containers, nitrogen flush se roka jaata hai. [Q1390, Q1401]
- Silver tarnishing: $2\text{Ag} + \text{H}_2\text{S} \rightarrow \text{Ag}_2\text{S (black)} + \text{H}_2$; corrosion ka example. [Q1370]
- Corrosion $\neq$ Rancidity: Corrosion = metal surface damage; Rancidity = fats/oils spoilage. [Q1390]

6. Isotopes, Isobars & Isomers

- Isotopes: Same chemical properties, different atomic masses; same atomic number.
 - Examples: C-12, C-13, C-14; H-1, D-2, T-3; U-235, U-238. [Q1379]
- Isobars: Same mass number, different atomic number.
 - Examples: ${}_{19}\text{K}^{40}$, ${}_{18}\text{Ar}^{40}$, ${}_{20}\text{Ca}^{40}$, ${}_{11}\text{Na}^{24}$, ${}_{12}\text{Mg}^{24}$. [Q1379]

- Isomers: Same chemical formula, different properties/structure.
 - Examples: Ethyl alcohol & dimethyl ether (C_2H_6O); Pentane, iso-pentane, neopentane. [Q1379]

7. States of Matter & Physical Properties

- Force of attraction: Solid > Liquid > Gas > Plasma. [Q1391]
- Solid: Closely packed particles; maximum intermolecular force. [Q1391]
- Liquid: Moderate force; particles close but can move. [Q1391]
- Gas: Very weak force; particles free to move. [Q1391]
- Plasma: 4th state of matter; no fixed shape/volume; less dense than solids/liquids; Sir William Crookes (1879) identified; Irving Langmuir (1928) described; fluorescent tube mein gas ki nature pe depend karke special colour glow karta hai. [Q1399]
- Bose-Einstein Condensate (BEC): Super cold atoms; Satyendra Nath Bose aur Albert Einstein ne 1924 mein predict kiya. [Q1399]
- Relative density: Density of substance / Density of water; unitless; water = 1 g/cm^3 at 4°C . [Q1392]
- Liquefaction & boiling points: Stronger intermolecular forces aur higher molecular weight $\rightarrow$ higher points. Acetic acid > Chloroform > Ethanol > Methane. [Q1385]
- Water molecule: H:O ratio = 2:1 (H_2O). [Q1388]
- Mg burning: Dazzling white flame; white ash (MgO) $\rightarrow$ water mein dissolve karke $Mg(OH)_2$ banata hai. [Q1400]

8. Fireworks, Flame Tests & Gas Colours

- Fireworks colours (metal salts):
 - Green: Barium (Ba, atomic number 56). [Q1373]
 - Blue: Copper (Cu). [Q1373]
 - Orange: Calcium (Ca). [Q1373]
 - Yellow: Sodium (Na). [Q1373]
 - Red: Lithium (Li). [Q1373]
 - Golden sparks: Iron filings (magnetite, hematite) + charcoal. [Q1373]
- Flame colours:
 - Blue flame: Complete combustion; sufficient oxygen. [Q1398]
 - Red/Orange/Green flame: Incomplete combustion. [Q1398]
- LPG components: Propane (C_3H_8), butane (C_4H_{10}), propylene (C_3H_6), butylene (C_4H_8), isobutane (C_4H_{10}). [Q1398]

9. Acid Rain & Environment

- Acid rain: pH $\sim$ 5.2 ya usse kam; SO_2 aur nitrogen oxides (NO , NO_2) se banta hai; fossil fuel combustion se emit hote hain.
 - $SO_2 + H_2O + O_2 \rightarrow H_2SO_4$ (sulphuric acid).
 - $NO_x + H_2O + O_2 \rightarrow HNO_3$ (nitric acid). [Q1371, Q1374]
- Acid rain se related gases: SO_2 , SO_3 , NO , NO_2 . [Q1371, Q1374]
- NH_3 acid rain se related NAHI hai. [Q1371]

10. Photography & Silver Compounds

- Black & white photography: AgCl aur AgBr use hote hain.
 - $2\text{AgCl} \rightarrow 2\text{Ag} + \text{Cl}_2$ (sunlight); $2\text{AgBr} \rightarrow 2\text{Ag} + \text{Br}_2$.
 - Light absorb hoti hai — isliye endothermic reaction.
 - Sodium thiosulfate ($\text{Na}_2\text{S}_2\text{O}_3$, hypo) se fix karte hain — unchanged silver salts dissolve ho jaate hain. [Q1389]
- AgCl: White; sunlight mein grey ho jaata hai (decomposition into Ag + Cl_2); dark bottle mein store karte hain; mercury poisoning ke liye antidote. [Q1357]
- AgBr: Same behaviour as AgCl. [Q1357]

11. Coal, Coal Tar & Fractional Distillation

- Coal tar: Thick black liquid, unpleasant smell; different carbon compounds ka mixture.
 - Fractional distillation se dyes, explosives, paints, synthetic fibres, drugs, pesticides milta hain. [Q1363]
- Naphthalene: Aromatic hydrocarbon; coal tar ya crude oil mein milta hai; moth repellent (moth balls); plastics, resins, fuels, dyes manufacturing mein. [Q1363]
- Bitumen: Coal tar se obtained; road construction, waterproofing mein use hota hai. [Q1363]

12. Chemical vs Physical Change

- Chemical change: New substance forms; mostly irreversible.
 - Examples: Rusting, digestion, baking a cake, rotting fruits, decomposition of waste, butter turning rancid. [Q1401]
- Physical change: No new substance; reversible.
 - Examples: $\text{CO}_2 \rightarrow$ dry ice, magnetisation of iron, heating platinum wire, boiling water, vaporizing liquid nitrogen. [Q1401]
- Iodine-Starch test: Potato juice mein starch hota hai — Iodine (I) starch $[(\text{C}_6\text{H}_{10}\text{O}_5)_n]$ ke saath react karke blue-black colour banata hai. Iodine photography, pharmaceutical industry, water purifiers mein bhi use hota hai. [Q1350]
- Enzyme test example: Test tube A mein starch solution + dilute iodine $\rightarrow$ blue-black colour. Test tube B mein starch + saliva — salivary amylase enzyme starch ko simple sugars mein todta hai, isliye iodine sugars ke saath complex nahi banata aur blue-black colour NAHI dikhta. [Q1362]

13. Electrolysis & Electrical Conduction

- Brine electrolysis (Chlor-alkali process):
 - Cathode: $2\text{H}_2\text{O} + 2\text{e}^- \rightarrow \text{H}_2\uparrow + 2\text{OH}^-$ (H_2 gas — pop sound).
 - Anode: $2\text{Cl}^- \rightarrow \text{Cl}_2\uparrow + 2\text{e}^-$.
 - Overall: $2\text{NaCl} + 2\text{H}_2\text{O} \rightarrow \text{H}_2 + \text{Cl}_2 + 2\text{NaOH}$.
 - Cathode = current flows out; anode = current enters. [Q1358]
- Sodium chloride (aq): Conducts electricity — dissociates into $\text{Na}^+ + \text{Cl}^-$. [Q1366]
- Sugar, glucose, alcohol, kerosene, octane: Do NOT conduct electricity — no free ions. [Q1366, Q1387, Q1404]

14. Water, Hardness & Solutions

- Hard water: >50-60 mg/L calcium/magnesium carbonate. [Q1396]
- Washing soda (Na_2CO_3): Temporary aur permanent dono hardness remove karta hai. [Q1396]
- Water of crystallization: Crystals heat karne pe lost hota hai $\rightarrow$ anhydrous form.
 - $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$ = washing soda crystals. [Q1351]
 - $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ (blue) $\rightarrow$ heat $\rightarrow$ anhydrous CuSO_4 (white). [Q1308]

15. Fire Extinguishers & Safety

- Soda-acid fire extinguisher: Sodium bicarbonate (NaHCO_3) + sulphuric acid — CO_2 gas evolved; fire ko suffocate karta hai.
 - ⚠ Trap: Sodium carbonate nahi — sodium bicarbonate use hota hai. [Q1351, Q1353]
- CO_2 fire extinguishers: CO_2 gas use hoti hai. [Q1353]

16. ⚠ Traps / Key Differences Summary

- ⚠ Trap: Soda-acid fire extinguisher mein NaHCO_3 use hota hai — Na_2CO_3 nahi. [Q1351]
- ⚠ Trap: NaOH permanent hardness remove karne ke liye use NAHI hota. [Q1354]
- ⚠ Trap: Covalent compounds ka high melting/boiling point property nahi hai — ye ionic ka hai. [Q1386]
- ⚠ Trap: Ionic compounds ka low melting/boiling point property nahi hai. [Q1397]
- ⚠ Trap: Sugar solution electricity nahi conduct karta — ions nahi hote. [Q1387]
- ⚠ Trap: Corrosion aur rancidity alag-alag cheezein hain — corrosion = metal damage; rancidity = fats/oils spoilage. [Q1390]
- ⚠ Trap: NH_3 acid rain se related NAHI hai. [Q1371]
- ⚠ Trap: Blue flame = complete combustion; red/orange/green = incomplete combustion. [Q1398]
- ⚠ Trap: POP ($\text{CaSO}_4 \cdot \frac{1}{2}\text{H}_2\text{O}$) gypsum se 373 K par banta hai — 473 K ya 573 K nahi. [Q1355]
- ⚠ Trap: AgCl sunlight mein grey ho jaata hai — black nahi. [Q1357]
- ⚠ Trap: Same group ke elements ki chemical properties similar hoti hain — atomic mass/radius/number nahi. [Q1394]
- ⚠ Trap: Water molecule mein $\text{H}:\text{O} = 2:1$ hai — $1:1$ ya $1:2$ nahi. [Q1388]
- ⚠ Trap: Mg burning se MgO banta hai; water mein dissolve karke $\text{Mg}(\text{OH})_2$ banata hai — direct $\text{Mg}(\text{OH})_2$ nahi. [Q1400]
- ⚠ Trap: Denatured alcohol mein methanol milaya jaata hai — safe nahi peena. [Q1395]
- ⚠ Trap: $\text{Pb}(\text{NO}_3)_2 + \text{KI}$ se banne wala precipitate YELLOW hota hai (PbI_2) — white nahi. [Q1356]